

## A Homework: Origins (due Thu 19th Feb 5pm)

### A.1 Particle physics units

Units:  $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ ,  $c = 3 \times 10^8 \text{ m s}^{-1}$ ,  $\hbar c = 197 \text{ MeV fm}$ ,  $\hbar = 1.05 \times 10^{-34} \text{ J s}$ .

(a) Convert the electron and proton mass is  $m_e = 9.11 \times 10^{-31} \text{ kg}$  and  $m_p = 1.67 \times 10^{-27} \text{ kg}$  into units of MeV. Numerically estimate the reduced Compton wavelengths  $\lambda_C = \hbar/mc$  of an electron, proton, and a Higgs boson  $m_h = 125 \text{ GeV}$ .

(b) Muons undergo beta decay at a rate given by  $\Gamma = (G_F^2 m_\mu^5)/(192\pi^3)$ . Given the Fermi constant is  $G_F = 1.17 \times 10^{-5} \text{ GeV}^{-2}$  and muon mass  $m_\mu = 106 \text{ MeV}$ , numerically calculate the rest lifetime of the muon  $\tau = \hbar/\Gamma$  in seconds.

(c) From the Lorentz force  $\frac{d\mathbf{p}}{dt} = q\mathbf{v} \times \mathbf{B}$  and centripetal acceleration  $a = v^2/R$ , derive

$$\left( \frac{p}{\text{GeV c}^{-1}} \right) = k \left( \frac{B}{1 \text{ T}} \right) \left( \frac{R}{1 \text{ m}} \right). \quad (\text{A.1})$$

relating a particle with unit charge  $q = 1.6 \times 10^{-19} \text{ C}$ , momentum  $p = \gamma mv$  at constant velocity perpendicular to the magnetic field  $B$ , bending radius  $R$ , and  $k \approx 0.3$ . Note that the SI units for Tesla are  $[B] = [f]/[qv] = \text{N}/(\text{C m s}^{-1})$ .

### A.2 Cosmic-ray relativistic kinematics

(a) In the lab frame, a cosmic ray with four-momentum  $P_C^\mu = (E, \mathbf{p})^T$  strikes a stationary nucleus with mass  $m_N$ . Show that the centre-of-mass energy is  $E_{\text{CM}} \approx \sqrt{2m_N E}$  and write the condition this holds. Use this to estimate the proton energy needed to strike a stationary oxygen nucleus ( $m_{\text{oxygen}} \approx 16m_{\text{proton}}$ ) with LHC energies:  $E_{\text{CM}} = 7$  and  $13.6 \text{ TeV}$ .

(b) In the “GZK process”, a proton strikes a primordial cosmic photon  $p + \gamma_{\text{cosmic}} \rightarrow n + \pi^+$ , producing a neutron and pion at threshold. Apply energy–momentum conservation to show the minimum proton energy is  $E_p \approx [(m_n + m_\pi)^2 - m_p^2]/4E_\gamma$ , and estimate its value given  $E_\gamma = 0.23 \text{ meV}$ ,  $m_p = 938.3 \text{ MeV}$ ,  $m_n = 939.6 \text{ MeV}$ ,  $m_\pi = 139.6 \text{ MeV}$ .

### A.3 Dirac equation

(a) From the Weyl representation of  $\gamma^\mu$  matrices eq. (3.50), write  $\gamma^0, \gamma^1, \gamma^2, \gamma^3$  as  $4 \times 4$  matrices. For  $\{\gamma^0, \gamma^1\}$  and  $\{\gamma^1, \gamma^1\}$ , show by explicit multiplication they satisfy  $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ .

(b) The Dirac equation in its coupled Weyl form is  $i\sigma^\mu \partial_\mu \chi = m\phi$  and  $i\bar{\sigma}^\mu \partial_\mu \phi = m\chi$ . Using the Pauli matrices (section 3.3) and definition of four-derivatives, write these as  $2 \times 2$  matrix equations in terms of energy  $E = i\partial_t$  and momentum  $\mathbf{p} = -i\nabla$  operators.